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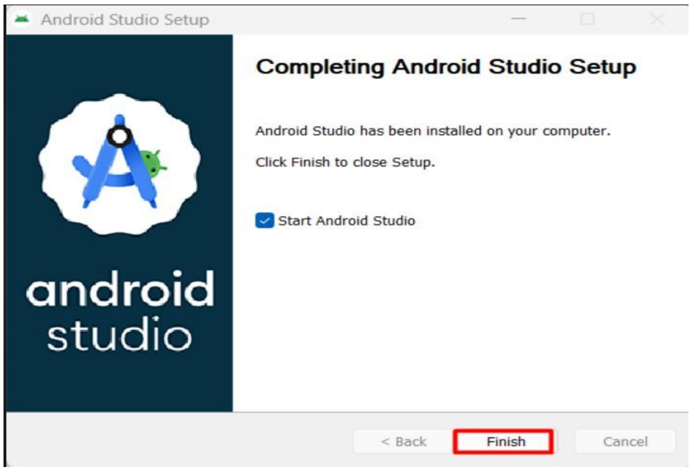
Experiment No.1

Aim: To install Android Studio and study its components.

Problem statement: Install Android Studio and study all components.

Questions:

1. **Install Android Studio.**



2. **Explain following components:**
   a. **Steps to create a new project in Android**
   To create a new Android project, open Android Studio and click on New Project. Select the Empty Activity or Empty Views Activity template and click Next. Enter the application name, package name, save location, programming language (Kotlin), and minimum SDK version. After entering the required details, click Finish. Android Studio will automatically create the project structure and open it for development.

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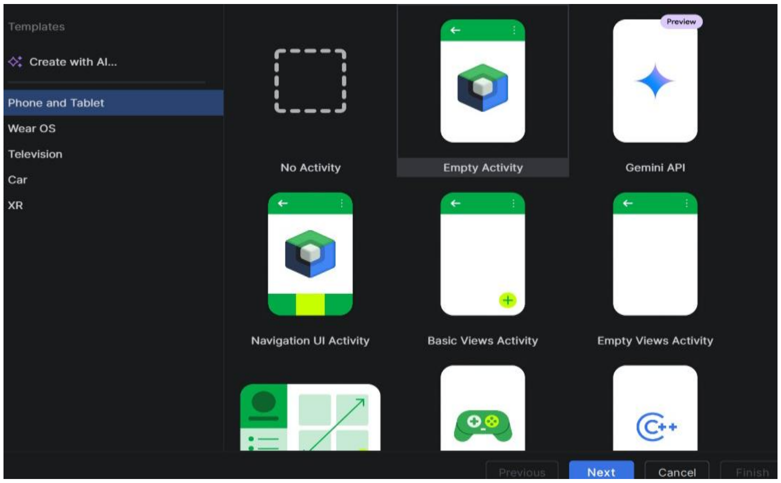
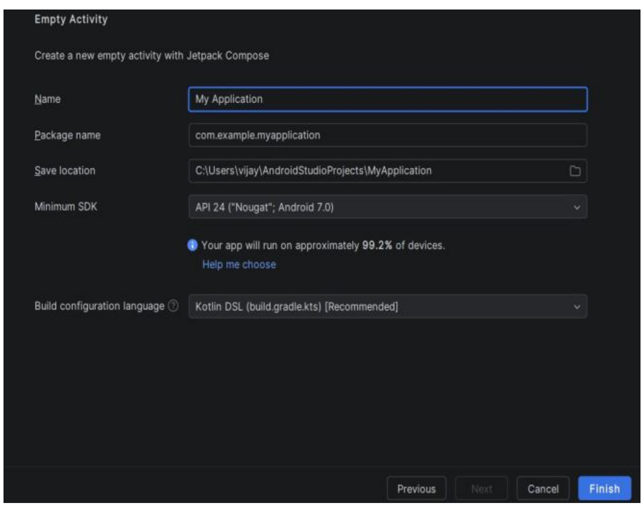


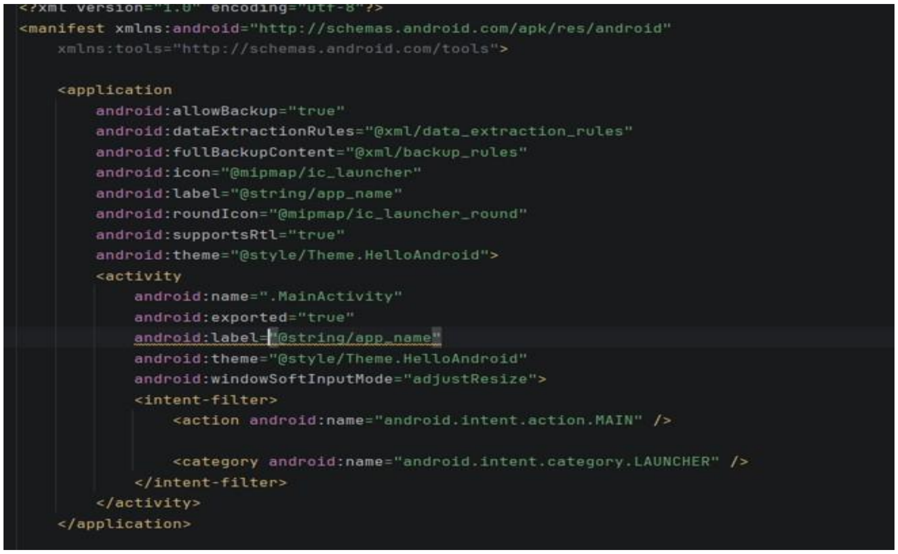
b.Activity (MainActivity.kt)

MainActivity.kt is the main Kotlin file of an Android application and represents the first screen displayed to the user. It contains the application logic and controls the user interface by connecting it with the layout file. The onCreate() method is executed when the activity starts. This file is responsible for handling user interactions and managing the activity lifecycle. Every Android application contains at least one activity.



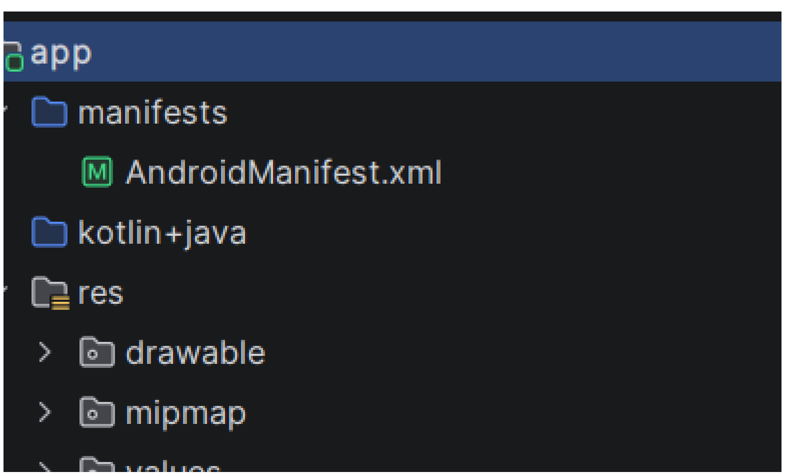
(b) activity_main.xml

The **activity_main.xml** file is the layout file used to design the graphical user interface (UI) of an Android application. It contains UI components such as TextView, Button, ImageView, EditText, and other widgets. The layout is written in XML, which separates the user interface from the application logic. This makes the application easier to design and maintain. The layout file is loaded by MainActivity.ktusing the setContentView()method.



C.Gradle Building Tool

Gradle is the build automation tool used in Android Studio to compile and package Android applications. It automatically downloads required libraries and manages project dependencies. Gradle also compiles the source code, generates the APK or App Bundle, and supports different build variants such as Debug and Release. It improves build performance and simplifies project management. Every Android project contains Gradle configuration files.



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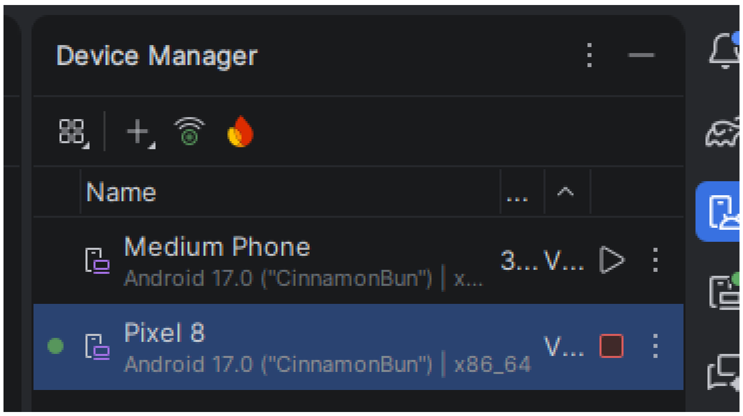
3. SDK and SDK Tools

The **Android Software Development Kit (SDK)** provides all the tools, libraries, and APIs required to develop Android applications. It includes the Android platform, build tools, platform tools, emulator, and debugging utilities. SDK Tools such as ADB, SDK Manager, and Logcat help developers build, test, and debug applications efficiently. The SDK is installed during Android Studio setup and can be updated whenever new Android versions are released.

4. AVD (Android Virtual Device)

An **Android Virtual Device (AVD)** is an emulator that simulates a real Android smartphone or tablet on a computer. It allows developers to test applications without using a physical device. Different AVDs can be created with various Android versions, screen sizes, and hardware configurations. It helps in testing application compatibility and debugging. The AVD is managed through the Device Manager in Android Studio.



5. Steps to Create a New Virtual Device

To create a virtual device, open Android Studio and select **Tools → Device Manager**. Click on **Create Device** and choose a hardware profile such as Pixel 6. Next, select an Android system image and download it if it is not already installed. Click **Next**, review the configuration, and then click **Finish**. Finally, start the virtual device by clicking the **Play** button, and it will launch the Android emulator ready for testing applications.

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